



Mining in space

A new investigation on the ISS tests BioRock, extraterrestrial biomineral by harnessing the power of microbes. <https://digit.in/aug19-20>



Apollo astronauts dealt with customs

Apollo 11 astronauts had to deal with customs after returning from the first Moon mission 50 years ago. <https://digit.in/aug19-19>

BREAKTHROUGH LISTEN

The VLT

Credit: ESO

Breakthrough Listen is an ambitious project spanning 10 years to use the most powerful astronomical instruments and computers for one purpose - finding signs of intelligent life in our stellar neighbourhood.

Aditya Madanapalle | aditya@digit.in

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reakthrough Listen is the biggest ongoing SETI project, and is using a number of astronomical

instruments over a period of ten years to detect for signs of advanced extraterrestrial civilisations. There is an ongoing radio and optical survey of 1,000,000 million stars closest to the Earth, the galactic plane of the Milky Way and the 100 nearest galaxies for signs of technologies deployed by advanced civilisations, also known as technosignatures.

Breakthrough Listen was co-founded by Stephen Hawking collaborates with a number of observatories around the world, including the Jodrell Bank Observatory, the Green Bank telescope and the CSIRO Parkes Radio Telescope. Apart from looking for technosignatures, the initiative also contributes to finding exoplanets, detecting fast radio bursts and studying interesting objects such as 'Oumuamua.

FAST RADIO BURSTS

Fast radio bursts (FRBs) are radio pulses that last anywhere between a

fraction of a millisecond all the way to a few milliseconds. Within this short period of time, FRBs release as much energy as the Sun does in an entire day. FRBs are made up of extremely high-energy radio waves, and are relatively new to science. The first FRB was observed only in the year 2007. Since then, several additional FRBs have also been detected. The process behind this and the exact sources of these FRBs are not yet fully understood. All the known FRBs are extragalactic in origin. Possible explanations for FRBs include cataclysmic deaths of the sources, neutron stars that are spinning at incredible speeds, a neutron star adjacent to a supermassive black hole, or advanced extraterrestrials using directed energy beams to power up their spacecraft. In 2015, a FRB was known to repeat, ruling out the destruction of the source explanation, at least for the repeating source.